DESIGN BRIEF · OCT 2026 · OWNER KR0

DUAL TECH

A short JRPG where every attack ends in a timing hit, and each hero's hit feels different. Brick holds and charges, Vex reacts, Lumen keeps the beat. Chain two heroes' hits and they fuse into a dual tech.

Genre JRPG, timed hits | Platform browser, keyboard, mouse or touch | Run 15 to 20 minutes | Budget 33.6M tokens

1 EXECUTIVE SUMMARY

Dual Tech is a five-fight run with three heroes. Turn bars fill in real time; when a hero's bar is full the fight pauses and you pick a move. Then you perform it with one button, and how well you time it decides the damage. Brick tests patience, Vex reflexes, Lumen rhythm. Good hits fill a shared Sync meter that unlocks dual techs, where two heroes chain their timing hits back to back. Beat the scrap golem in fight five to finish the run.

PLAYER FANTASY	WIN	LOSE
Landing a perfect two-hero combo on a boss, by hand.	Defeat the scrap golem in the fifth fight. You get a score from perfects, time and damage taken.	All three heroes are down. The run ends, you see your score so far, and a retry starts from fight 1.
THIS SEMESTER SHIPS 3 heroes, 6 abilities, 4 dual techs, 5 encounters and 15 upgrades.		

One hero, one feel

Each hero's timing style is a different skill: patience, reflex, rhythm. You should be able to tell who's acting with your eyes closed.

Decide, then perform

The menu is the strategy, and the hit is a one-to-two-second burst of skill. Neither half works alone.

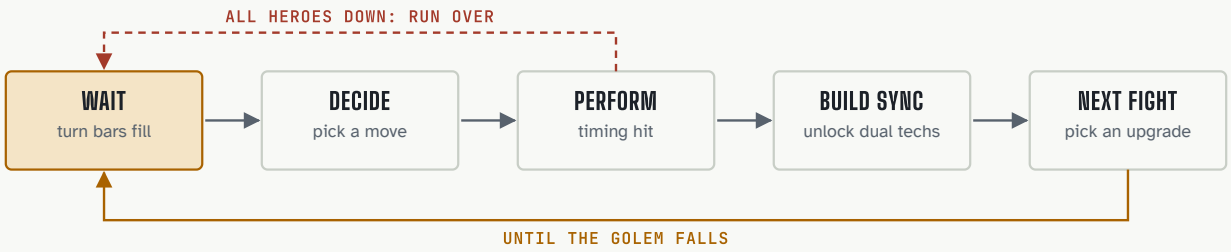
Combos are the payoff

Dual techs chain two feels back to back. Landing both perfectly is the best moment in the game.

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2 GAME MECHANICS

Core loop



A turn takes about 5 seconds. After each of the first four fights you pick 1 of 3 upgrades; HP carries over and Sync resets.

1. **Wait.** Turn bars fill in real time; the enemy's fills too.
2. **Decide.** A hero's bar is full, the fight pauses, and you pick an ability or a dual tech and a target.
3. **Perform.** Play that hero's timing hit. The grade, Perfect, Good or Miss, sets the effect.
4. **Build Sync.** Good and Perfect hits fill the shared meter; at 50 you can open a dual tech.
5. **Repeat** until the enemy or the party falls, then pick an upgrade and move to the next fight.

How you win and lose

WIN THE RUN

Clear four fights and defeat the 700-HP scrap golem. The results screen shows your score: 100 per perfect, plus time under 20 minutes, minus damage taken.

LOSE THE RUN

Every hero is down. You see the results screen with your score so far, and a retry starts again from fight 1 with no upgrades.

Three heroes, three feels

HERO	WHAT YOU DO	WHAT YOU SEE AND HEAR
Brick, tank	Hold to fill a charge bar and let go inside the zone. Hold too long and it overcharges into a Miss.	The bar fills, the screen tightens, a low hum rises; a perfect slam lands with hit-stop and heavy shake.
Vex, damage	A marker sweeps across a bar; press to stop it in a random zone. Flurry's three sweeps get faster.	A thin bright marker and a sharp click on each sweep; a perfect hit cracks and flashes white.
Lumen, support	Beats travel toward a ring 520 ms apart; tap as each one crosses.	Soft chimes on the music's tempo; a chord builds with each perfect beat.

Every hit uses the same button, Space, a click or a tap, so what changes from hero to hero is the skill, not the controls. Keys 1–4 pick a command.

The golem fight, turn by turn

Turn 1	Vex's red bar fills first. You pick Railshot; a marker sweeps across the bar and you stop it dead in the zone: **Perfect**, a white flash and double damage. Sync jumps by 12.
Turn 2	The golem's bar fills and it slams Lumen. You answer with Lumen's Pulse heal: three beats glide toward the ring and you tap all three on the chime. Everyone heals 36.
Turn 4	Sync passes 50 and the meter lights up. Brick opens **Ricochet slam**: you hold Brick's charge and release in the zone, then without a pause Vex's marker starts sweeping. You stop it in the zone. A name banner, a moment of slow motion, and the golem is staggered.

Turn 7

Dual techs

At half health the golem overheats and glows. Its bar fills faster and it hits 35% harder. You switch to Brick's Barrier wall to pull its attacks onto him.

A dual tech costs 50 Sync, empties the opener's bar and half of the partner's. This semester builds four: **Ricochet slam** (Brick's hold, then Vex's sweep), **Bulwark hymn** (Lumen's beats, then Brick's hold, healing and shielding the party), **Prism shot** (Lumen's beats, then Vex's sweep, hitting every enemy) and **Breach** (Brick's slam, then three Flurry sweeps).

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3 AI ARCHITECTURE

Each of the five encounters is an in-game agent with its own rule for when it acts, who it targets and one twist that tests one hero's style. You read every one of them from its turn bar. All run on fixed rules in the game, with no language-model calls during a run.

Rust rats ×3

WHAT YOU SEE

Three weak, fast rats whose bars fill quickly; each bites a random hero for 8.

EFFECT ON PLAY

The tutorial fight. Vex's Flurry sweeps jump to the next rat when one dies, so it clears the pack.

Bell warden

WHAT YOU SEE

Always hits your lowest-HP hero. Every third action it rings its bell instead, and Lumen's beats speed up to double time for two turns.

EFFECT ON PLAY

Rhythm under pressure: you have to keep Lumen healing while the beat doubles.

Plated beetle

WHAT YOU SEE

Good and Miss hits clang off its armor for 0 damage. Only Perfect hits get through.

EFFECT ON PLAY

Brick's perfect Shield slam staggers it and resets its bar, buying Vex a clean shot.

Twin drones

WHAT YOU SEE

Two drones with separate bars. A drone you haven't hit since its last turn heals its partner for 25 instead of attacking.

EFFECT ON PLAY

Hitting one drone at a time never wins. Prism shot, the dual tech that hits every enemy, is the answer.

Scrap golem (boss)

WHAT YOU SEE

A 700-HP golem that slams a random hero. At half health it overheats: its bar fills faster and it hits 35% harder.

EFFECT ON PLAY

The final exam. You need Barrier wall's taunt, Lumen's heals and big dual techs together.

How the agents play off each other. The twin drones are the one fight where two enemies cooperate. Each drone watches whether you hit it since its last turn, so the pair punishes focus fire. That's what teaches you to save Sync for a dual tech instead of spending every turn on single hits.

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4 TECHNICAL STRATEGY

Development agents

Twelve AI agents build Dual Tech. kr0 directs them and approves every design decision and every file they write.

AGENT	ROLE ON DUAL TECH
`creative-director`	Holds every hero, upgrade and dual tech to "one hero, one feel".
`technical-director`	Designed the single timing engine every hit runs through, and signs off architecture decisions.
`producer`	Runs the four milestones, tracks spend against the 33.6M-token budget, and applies the cut list in order if a milestone overruns.
`game-designer`	Owns the GDD: heroes, abilities, dual techs, encounters and upgrades.
`systems-designer`	Keeps the timing windows, grade multipliers and Sync costs in one tuning table.

AGENT	ROLE ON DUAL TECH
gameplay-programmer	Builds the battle, the abilities and each dual tech as a list of timing steps.
ai-programmer	Builds each enemy's targeting rule and twist: the bell, the beetle's armor, the drones' healing.
ui-programmer	Builds the hero panel, command menu, Sync meter and timing meter for keyboard and touch.
sound-designer	Makes each hero's sounds and keeps Lumen's beats locked to the 115 BPM battle track.
art-director	Keeps the pixel sprites and one colour per hero consistent across bars, sprites and pop-ups.
accessibility-specialist	Adds hero shapes for colour-blind players, the timing assist, the hold alternative and latency calibration.
qa-tester	Runs simulated-input fights at 10× speed, tests every timing window at its edges, and hand-plays each story.

How they hand work over. A dual tech is one data record: the game-designer writes its rule, the systems-designer adds its windows to the tuning table, the gameplay-programmer adds the record, and the qa-tester's input bot plays it at Perfect, Good and Miss before it's closed. When a milestone runs over, the producer cuts from the bottom of the cut list rather than squeezing every feature.

Semester scope

Assumes a 15-week semester ending in one complete run.

Ships this semester

- Core combat: 3 heroes, 6 abilities, Sync and Amp
- Ricochet slam, Bulwark hymn, Prism shot and Breach
- 5 encounters, upgrade picks and run scoring

Cut list, first to go

- Holy charge and Overclock volley (two inputs at once)
- The golem's laser phase
- Unlocks that carry between runs
- A mobile app build

- One battle track at 115 BPM, title and results screens
- Accessibility: hero shapes, timing assist, calibration

Constraints

Input timing

Lumen's perfect window is 70 ms either side of the beat, and Brick's is 38 ms. A dropped frame or device lag could turn a Perfect into a Good. So hits are graded by the input's timestamp, beats run on the audio clock, and players get a calibration screen.

No model calls during play

A hosted-model call takes far longer than a 70 ms window and costs tokens every run, so enemies act on fixed turn-bar rules: zero runtime tokens per run.

Content as data

Heroes, enemies, dual techs and upgrades are data records, so adding one is a short agent task. Only the cut dual techs with simultaneous inputs would need new code.

API constraints on the agent crew

- **Context window.** Agents load the GDD one section at a time; the tuning table is the most-read section.
- **Rate limits.** At most three agents work in parallel, and the timing engine has one owner at a time.
- **Model tiers.** Directors run on the strongest model; specialists run on the session's model.

Token budget

Planning estimate for the semester build. Most tokens are input, because every call re-reads the tuning table and code it touches. Actual spend per story goes into the tuning history and the plan is redone at each milestone gate.

FEATURE	MAIN AGENTS	TOKENS
Design upkeep and stories	game-designer, producer	1.5M

FEATURE	MAIN AGENTS	TOKENS
Timing engine and three hit styles	gameplay-programmer, technical-director	3.0M
Battle flow and turn bars	gameplay-programmer	2.0M
Six abilities, Sync and Amp	gameplay-programmer, systems-designer	2.5M
Four dual techs	gameplay-programmer	3.0M
Five encounters and enemy rules	ai-programmer	3.0M
Upgrades and run scoring	game-designer, gameplay-programmer	2.0M
Screens, HUD and accessibility	ui-programmer, accessibility-specialist	2.5M
Music sync and latency calibration	sound-designer	2.0M
Input bot, balance runs, playtests	qa-tester	2.5M
Subtotal		24.0M
Rework reserve (40%)		9.6M
Total development budget		33.6M
Runtime tokens per run		0